

Instructions

- Answer all eight questions. Show sufficient working for method marks.
- A calculator may be used unless a question states otherwise.
- Write clearly in the spaces provided. Additional pages may be attached.

1

Calculate the energy released when a body of mass 2.0 kg falls through 15 m, stating any assumption.

[4 marks]

$$E = mgh$$

2

Describe an experiment to measure the acceleration due to gravity, identifying the main source of error.

[6 marks]

3

A 60 kg skater pushes off a 40 kg skater and moves at 2 m/s. Find the other skater's speed.

[5 marks]

4

Explain the difference between distance and displacement using a closed-loop journey.

[4 marks]

5

Calculate the current through a $12\ \Omega$ resistor connected across a 9 V supply, and the power dissipated.

[5 marks]

$$V = IR$$

6

Describe how refraction changes wavelength but not frequency, and explain why.

[5 marks]

7

A wave has frequency 250 Hz and wavelength 1.4 m. Calculate its speed and state the equation used.

[4 marks]

$$v = f\lambda$$

8

Explain why sound cannot be polarised but light can, referring to the oscillation direction in each case.

[5 marks]
